

# Questions in DSA Journey

## DAY-1

### Basic Questions---

- 1) Tell lower case or upper case
- 2) Sum of n Natural numbers
- 3) Sum of odd numbers
- 4) Check number is Prime or Not
- 5) Average of marks
- 6) Check number is even or odd

### Variable and Data Types---

- 7) Program that converts a temperature from Celsius to Fahrenheit and vice versa.
- 8) Write a program that declares variables of different data types and prints their values.
- 9) Write a program to swap two variables without using a third variable
- 10) Program to calculate the area and circumference of a circle given the radius

### Conditional Statements---

- 11) Check number is positive, negative or zero using conditional statement
- 12) Find the Largest of three numbers
- 13) Check that age is eligible to vote
- 14) Check given year is leap or not
- 15) Make a basic calculator

### Loops---

- 16) Print the Table
- 17) Calculate the sum of all even number 1 to 100
- 18) Check prime using for loop
- 19) Factorial of a number using loop
- 20) Reverse a number using loop

### Functions---

- 21) Sum of two numbers using function
- 22) Min of two numbers in function
- 23) Sum of n numbers
- 24) Factorial of number
- 25) Sum of the digits of number
- 26) Calculate binomial coefficient  $nCr$
- 27) Write a function to print prime numbers 2 to n
- 28) Write a function to print nth Fibonacci
- 29) Write a function to print sum of Fibonacci numbers

### Bitwise Operators---

- 30) Convert decimal to Binary and vice versa
- 31) figure out how to find if a number is power of two
- 32) Reverse Integer

### DAY-5

#### Arrays---

- 33) Print Array values
- 34) Find the Smallest and Largest number in array and Those Index
- 35) Search the index of target value
- 36) Reverse Array
- 37) Write a function to calculate sum and product of Array
- 38) Write a function to swap max & min of an Array
- 39) Write a Function to print intersection of Two Array
- 40) Print Unique values
- 41) Vectors Initialization, For each loop

### DAY-7

#### Algorithms---

- 42) Maximum Sub-Array sum by Brute force
- 43) Brute force optimal
- 44) Kadane's Algorithm to calculate maximum sub-array sum
- 45) Two Sum = return indices of the two numbers such that they add up to target.
- 46) Pair sum = Return pair in sorted array with target sum = Brute force and 2 Pointer
- 47) Majority Element with brute force and optimal approach
- 48) Moore's Voting Algorithm to calculate Majority Element
- 49) Power Exponentiation ( $\text{Pow } x^n = O(n)$ ) and Binary Exponentiation =  $O(\log n)$
- 50) Buy and Sell stocks =  $O(n)$

#### Binary Search---

- 51) Search in rotated sorted Array : Leetcode 33
- 52) Peak Index in Mountain Array : Leetcode 852
- 53) Single Element in a sorted Array : Leetcode 540
- 54) Book Allocation
- 55) Painter's Partition Problem
- 56) Aggressive cows Problem

#### Sorting---

- 57) Bubble Sort
- 58) Selection Sort
- 59) Insertion Sort
- 60) Sort Array
- 61) Dutch National Flag Algorithm: Sort Array with 0s 1s & 2s : Time Complexity  $O(n)$
- 62) Merge two Sorted Array
- 63) Next Permutation

### Character Arrays and Strings---

- 64) Character Array: calculate length, Reverse and check palindrome
- 65) Valid Palindrome : Leetcode 125
- 66) Reverse words in a string 2 : Leetcode 186 medium(premium)
- 67) Maximum occurrence
- 68) Remove all adjacent duplicates in string : Leetcode 1047
- 69) string compression : Leetcode 443 (medium)

### 2D Arrays---

- 70) Initializing, taking Input and Printing 2D Arrays
- 71) Check target exist
- 72) Printing the sum
- 73) Largest Sum and rows/col index
- 74) Print like wave
- 75) Spiral Matrix : Leetcode 54 (Medium)
- 76) Rotate Image : Leetcode 48 (medium)
- 77) Search a 2D Matrix for sorted : Leetcode 74 (medium)
- 78) Search a 2D Matrix II : Leetcode 240 (medium)

### Maths---

- 79) Check Prime
- 80) Count Primes : Leetcode 204 (medium)
- 81) find GCD : gcd is like HCF
- 82) Modular Exponentiation : using fast exponentiation method

### Recursion---

- 83) Factorial in Recursion
- 84) Printing given number n to 1
- 85) Program that Tells Reached at home or not: two variables: step and dest
- 86) Fibonacci Number : Leetcode 509
- 87) Count ways to Reach the Nth stairs (Medium level)
- 88) Say Digits: Write a program that write Digits name
- 89) Calculate sum
- 90) Check sorted or not
- 91) Finding target in Recursion by Linear Search
- 92) Finding key in Recursion by Binary Search
- 93) String Reverse in Recursion
- 94) Check Palindrome
- 95) Power of  $a^b$  by Recursion
- 96) Bubble Sort in Recursion
- 97) Merge Sort
- 98) Quick Sort
- 99) Subsets : set of subsets : Leetcode 78 (medium)
- 100) Subsequence of String
- 101) Letter combination of a phone number : Leetcode 17 (medium) : Asked in MNCs
- 102) Permutations : Permutation of number : Leetcode 46 (medium)
- 103) Permutations : Permutation of string

- 104) Rat in a Maze Problem - I : GeeksForGeeks (medium) : Amazon, Microsoft

#### Linked List--- After Learning OOPs

- 105) Define the Structure of Linked List Node
- 106) Insertion / Deletion / Updation in Singly, Doubly and Circular Linked List
- 107) Reverse Linked List: Iteratively, Recursively and more Optimized in Recursion
- 108) Reverse a Linked List: Leetcode 206
- 109) Middle of Linked List: Leetcode 876
- 110) Reverse Linked List in k groups : Reverse nodes in k group: Leetcode 25(Hard)
- 111) Check list is circular or not
- 112) Detect a Loop: Linked List Cycle: Leetcode 141
- 113) Floyds Algo to detect a loop
- 114) Start node of loop
- 115) Remove loop from the list
- 116) Remove duplicates from Sorted list : Leetcode 83(easy)
- 117) Remove duplicates from Unsorted list
- 118) Break a circular linked list in two circular linked list
- 119) Sort 0s, 1s and 2s in Linked List: Sort Linked List of 0s 1s 2s : using link replacing
- 120) Merge two sorted List (Iterative & Recursive) Approach
- 121) Check if the linked list is Palindrome or Not
- 122) Add two numbers represented by Linked List: means sum of two linked list
- 123) Clone a linked list with next and random pointer
- 124) Merge Sort linked list, using links, without data/val replacement
- 125) Flatten a linked list

#### Stack---

- 126) Input & Output in Stack
- 127) Implementing stack (using Arrays & Linked List)
- 128) Two stack Implementation in a single Array
- 129) Reverse a string using stack
- 130) Delete middle element from the stack
- 131) Valid Parentheses
- 132) Insert an element at its bottom in a given stack
- 133) Reverse a stack using Recursion
- 134) Sort a Stack
- 135) Redundant brackets
- 136) Minimum cost to make string valid
- 137) Next smaller element: for input(2, 1, 4, 3) next smaller element is (1, -1, 3, -1)
- 138) Prev smaller element
- 139) Largest Rectangle in Histogram : Leetcode 84(Hard)
- 140) The celebrity problem
- 141) Max Rectangle in binary matrix with all 1's
- 142) N stacks in an Array
- 143) Design a stack that supports getMin() in O(1) time and O(1) extra space

### Queue---

- 144) Implementation of Queue using STL, Arrays, Linked List
- 145) Circular Queue
- 146) Doubly ended queue
- 147) Reverse queue using stack and recursion
- 148) First negative integer in every window of size k
- 149) Reverse first k element of queue
- 150) First non-repeating character in a string
- 151) Circular tour
- 152) Interleave the first half of queue with second half
- 153) Implement Queue using Stack, Implement Stack using Queue
- 154) K queues in an Array
- 155) Sum of minimum and maximum elements of all Sub-Arrays of size k

### Binary Tree---

- 156) Binary Tree Implementation
- 157) Build Tree, Build from level order
- 158) pre order Traversal Recursively and Iteratively
- 159) In order Traversal Recursively and Iteratively
- 160) post order Traversal Recursively and Iteratively
- 161) Level order Traversal
- 162) Count leaf nodes
- 163) Height of Binary Tree
- 164) Diameter of a Binary Tree: Longest path between any two end nodes
- 165) Check for balance tree, and optimized approach
- 166) Determine if two tree are identical
- 167) Sum Tree
- 168) Zig Zag Traversal
- 169) Boundary Traversal of Binary Tree
- 170) Vertical Traversal of Binary Tree
- 171) Top view of a Binary Tree: Bottom view of binary tree
- 172) Left view of a Binary Tree: Right view of Binary Tree
- 173) Diagonal Traversal of Binary Tree: check out geeksforgeeks for all Tree Problems
- 174) Sum of the Longest Bloodline in a tree: sum of nodes of longest path in tree
- 175) Lowest common Ancestor in a binary tree
- 176) K sum paths
- 177) Kth Ancestor in a Tree
- 178) Maximum sum of Non-Adjacent Nodes
- 179) Construct Tree from In order & Preorder
- 180) Tree from post order and in order
- 181) Problem: Burning Tree(Hard) : min time to burn a tree
- 182) Moris Traversal
- 183) Flatten binary tree to linked list

#### Binary Search Tree----

- 184) BST Implementation: Insertion, Deletion and Searching
- 185) Find min and max val of BST
- 186) Inorder Predecessor(prev) and Successor(next)
- 187) Validate BST
- 188) Find Kth smallest element in BST
- 189) Find Kth Largest element in BST
- 190) Find Kth Predecessor and successor
- 191) LCA in BST : Least/Lowest common ancestor of two nodes
- 192) Two sum in BST
- 193) Flatten BST to a sorted list
- 194) Normal BST to balanced BST
- 195) Preorder Traversal of a BST
- 196) Merge Two BSTs: and make one
- 197) Largest BST in a Binary Tree : return size(number of nodes)

#### Heaps---

- 198) Heap Implementation: and Insertion : Deletion : Heap Sort
- 199) Heapify Algo : Min Heap and Max Heap
- 200) Priority Queue
- 201) Kth smallest element : For kth largest element we use min heap
- 202) Is Binary Tree Heap
- 203) Merge two Binary Max-Heap
- 204) Min cost of Ropes
- 205) Convert BST to Min Heap
- 206) kth largest subarray sum
- 207) Merge k sorted Arrays
- 208) Smallest range from k sorted list
- 209) Median in a Stream (Hard)

#### HashMap---

- 210) Implementation of HashMap/Hashing
- 211) Maximum frequency number

#### Trie---

- 212) Trie Implementation: insertion/deletion/searching =  $O(l)$ ,  $l$  is length of word
- 213) Lowest common prefix
- 214) Implement a phone directory

### Backtracking---

- 215) Rat in a maze
- 216) N Queen Problem
- 217) Sudoku solver

### Graph---

- 218) Graph Implementation
- 219) BFS Traversal
- 220) DFS Traversal
- 221) Detect cycle in a undirected graph by **\*\*BFS\*\***
- 222) Detect cycle in a undirected graph by **\*\*DFS\*\***
- 223) Detect cycle in a **\*\*\*Directed graph by \*\*DFS\*\***
- 224) Topological sort using DFS : use on only directed acyclic graph(DAG)
- 225) Topological sort : Kahn's Algorithm : DAG : using BFS
- 226) Cycle detection in Directed graph : kahn's algo bfs
- 227) Shortest Path in DAG
- 228) Shortest path in Undirected Graph
- 229) Dijkstra's Algorithm | to find shortest path | using set | using priority queue
- 230) Minimum spanning tree | using Prime's Algo
- 231) DISJOINT SET : Minimum spanning tree | using Kruskal's Algo
- 232) Bridge in Graph
- 233) Critical connections in a network: Leetcode-1192 (Hard)
- 234) Articulation points in a Graph
- 235) Kosaraju's Algorithm : Find Strongly connected components(SCCs)
- 236) Bellman Ford Algorithm : To find shortest path : it can work on negative weight

### Dynamic Programming----

- 237) Dynamic Programming Recursion | Memoization | Tabulation | Space Optimize
- 238) Fibonacci of n
- 239) Minimum cost to climbing stairs: Leetcode 70 & Leetcode 746
- 240) Minimum Number of Coins to be Added(Medium) or Min coin | Leetcode(2952)
- 241) Maximum sum of non adjacent elements
- 242) House Robbery Problem
- 243) Cut Rod into segments of x y x : Rod cutting Problem
- 244) Count Derangements
- 245) Painting Fence Algorithm
- 246) 0/1 KnapSack Problem
- 247) Combination Sum IV(Leetcode-377)
- 248) Get Minimum Squares
- 249) Minimum Costs For Tickets(Leetcode-983)
- 250) Largest square formed(area) in a matrix
- 251) Leetcode(1039): Minimum Score Triangulation of Polygon
- 252) Leetcode(1824): Minimum Sideway Jumps
- 253) Leetcode(1402): Reducing dishes(hard)

- 254) Largest increasing subsequence
- 255) Leetcode(354): Russian Doll Envelopes(Hard)
- 256) Leetcode(1691): Maximum Height by Stacking Cuboids(Hard)
- 257) Leetcode(1388): Pizza with 3n Slices
- 258) Number of Dice Rolls with Target Sum || Dice Rolls
- 259) Leetcode(416): Partition Equal Subset Sum
- 260) Leetcode(801): Minimum Swaps to make Subsequences Increasing(Hard)
- 261) Leetcode(1027): Longest Arithmetic Subsequence/Progresssion
- 262) Leetcode(1218): Longest AP with given Difference
- 263) Leetcode(96): Unique Binary Search Trees
- 264) Leetcode(375): Guess Number Higher or Lower II
- 265) Leetcode(1130): Minimum Cost Tree From Leaf Values
- 266) Leetcode(121): Best time to buy and sell stocks
- 267) Leetcode(122): Best time to buy and sell stocks II
- 268) Leetcode(123): Best Time to Buy and Sell Stock III
- 269) Leetcode(188): Best Time to Buy and Sell Stock IV
- 270) Leetcode(714): Best Time to Buy and Sell Stock with Transaction Fee
- 271) Leetcode(1143): Longest common subsequence
- 272) Leetcode(516): Longest palindrome subsequence
- 273) Leetcode(72) : Edit Distance (Hard)
- 274) Leetcode(85): Maximal Rectangle
- 275) Leetcode(44): Wildcard Matching

#### Greedy Algorithm----

- 276) N meetings in one room
- 277) Maximum meetings in one room
- 278) Shop in candy store
- 279) Check if it is possible to survive on Island
- 280) Reverse Words
- 281) Chocolate distribution problem
- 282) Min cost of ropes
- 283) Huffman Encoding
- 284) Fractional knapsack
- 285) Activity selction problem
- 286) Job sequencing problem